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# Import libraries
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get uploaded image name
image_path = next(iter(uploaded))

# Read image
img = cv2.imread(image_path)

# Check if image is loaded
if img is None:
    print("Error: Image not loaded!")
else:
    # Get image dimensions
    rows, cols = img.shape[:2]

    # Source points (Corners of the original image)
    src_points = np.float32([
        [0, 0],
        [cols - 1, 0],
        [0, rows - 1],
        [cols - 1, rows - 1]
    ])

    # Destination points (Apply DLT Transformation)
    dst_points = np.float32([
        [50, 50],
        [cols - 100, 20],
        [20, rows - 50],
        [cols - 50, rows - 20]
    ])

    # Compute Homography Matrix using Direct Linear Transformation (DLT)
    H, status = cv2.findHomography(src_points, dst_points, method=0)

    # Apply the transformation
    transformed_img = cv2.warpPerspective(img, H, (cols, rows))

    # Display images
    print("Original Image")
    cv2_imshow(img)

    print("Image after Direct Linear Transformation (DLT)")
    cv2_imshow(transformed_img)
```

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cv2.imshow('transformed_img')

# Save the transformed image
cv2.imwrite("DLT_Transformed_Image.jpg", transformed_img)

# Download the output image
files.download("DLT_Transformed_Image.jpg")
```

Choose Files CVVV4.png

CVVV4.png(image/png) - 463386 bytes, last modified: 7/22/2026 - 100% done

Saving CVVV4.png to CVVV4.png

Original Image



Image after Direct Linear Transformation (DLT)

